

# MANNAT PAL

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## EDUCATION

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Vellore Institute of Technology

Integrated M.Tech – Computer Science and Engineering, Cumulative GPA: 8.89

Bhopal, M.P.

Oct. 2022 - Present

MASD Public School

Class XII, 86.4%

Panipat, HR

March 2021

Airforce School

Class X, 89.4%

Gorakhpur, U.P.

April 2019

## SKILLS

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**Security Tools:** Burp Suite, Metasploit, Nmap, Wireshark, SQLMap, Nikto, OWASP ZAP, Androguard, APKTool

**Machine Learning:** Scikit-learn (Random Forest, XGBoost, Naive Bayes), Pandas, NumPy, Matplotlib

**Programming & Databases:** C++, Python, HTML, CSS, SQL

**Platforms & Developer Tools:** Git, GitHub, Docker, AWS, VS Code, Vercel

**Computer Science Fundamentals:** Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

## EXPERIENCE

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Cybersecurity Analyst Intern

Nov 2025 - Dec 2025

Sequaretek IT Solutions Pvt. Ltd.

Mumbai, India

- Developed a **behavior-based malware analysis** pipeline using honeypot telemetry and large language models, achieving **74% detection accuracy**.
- **Implemented automated pre-analysis** modules to extract behavioral indicators, file hashes, and severity scores for malware classification.
- **Generated structured malware analysis reports**, indicators of compromise (IoCs), and threat intelligence summaries using **open-source** large language models such as **GPT-OSS and Mistral**.
- Conducted **threat intelligence validation** using VirusTotal and hash-based reputation analysis.

## PROJECTS

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Malware Analysis for Android APK using Machine Learning ([Project Link](#))

- Conducted **static analysis on 429 Android APKs** by extracting manifest-level features such as permissions, intents, and component metadata using **Python and Androguard**.
- **Trained and evaluated Random Forest, XGBoost, and ensemble machine learning** models for multi-class Android malware detection, achieving **92.86% accuracy and a 90.48% macro F1-score**.
- **Built an imbalance-aware malware triage pipeline** to reduce false positives and systematically prioritize high-risk applications for **analyst-driven investigation**.

Encryption & Decryption Algorithms ([Project Link](#))

- **Implemented and optimized** more than 10 cryptographic algorithms in **Python**, including RSA, Diffie-Hellman, AES, IDEA, HMAC, and classical cipher techniques.
- **Improved cryptographic performance by 30%** through efficient key generation strategies, modular algorithm design, and targeted code refactoring.
- Validated encryption, decryption, and data integrity workflows across all implementations, **achieving 99% functional accuracy and reliable reversibility**.

## EXTRA-CURRICULAR ACTIVITIES

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- **Event Management Lead**, Mozilla Firefox Club – Organized 5+ technical events.
- **Content and Editorial Lead**, Cyber Warriors Club – Led cybersecurity awareness initiatives.
- **Class Representative** – Coordinated academic and administrative resolution for 70+ students.